



Graduation Project Proposal Title

By

Ahmed Arafa Hafez – 2201055

Mohamed Bahaa Eldien Hamouda – 2201019

Manar Alaa Ahmed – 2200976

Mahmoud Abdellaal Mahmoud – 2201058

Monica Basem Wadea – 2200955

Ahmed Kamal Ragb – 2201301

Ahmed Hossam Abdelwahab – 2100332

Karim Tarek Ali – 2100835

Supervisor Eng/

Sara Hamdy

The Egyptian E-Learning University

Faculty of Computers and Information Technology

BSc

June 2026

Abstract

This project aims to develop an integrated safety and management system for industrial environments, leveraging Artificial Intelligence and Internet of Things technologies. The system automatically monitors events within the factory, such as detecting conflicts or violent incidents and sending instant alerts to security. It also ensures compliance with safety measures by detecting whether employees are wearing protective helmets. Additionally, the system provides attendance and departure tracking through facial recognition to enhance accuracy and security. Furthermore, the project utilizes sensor data to predict potential machinery failures, helping to minimize downtime and improve operational efficiency. This solution seeks to foster a safer workplace and increase productivity by combining AI-driven monitoring with predictive analytics

1 Introduction

Modern factories face multiple challenges related to safety and productivity, where any lapse in safety measures or unexpected equipment failures may lead to significant human and financial losses. With the advancement of Artificial Intelligence (AI) and the Internet of Things (IoT), it is now possible to design smart systems capable of monitoring the work environment in real time and responding proactively to potential risks.

This project focuses on developing an integrated system that enhances workplace safety by detecting undesirable behaviors such as conflicts, verifying employee compliance with protective measures like helmet usage, and ensuring accurate attendance management through facial recognition technology. In addition, the system leverages predictive analytics based on sensor data to forecast potential machinery failures before they occur, thus improving operational efficiency and minimizing downtime. Therefore, the project aims to combine safety and efficiency within a technologically advanced framework tailored to industrial environments.

1.1 Problem Statement

Many factories face recurring challenges related to employee safety and operational efficiency. Conflicts or undesirable behaviors in the workplace can disrupt production flow and pose security risks.

Furthermore, the lack of compliance with safety measures, such as wearing protective helmets, increases the likelihood of accidents and injuries. Traditional attendance systems are often prone to manipulation and errors, reducing their reliability. In addition, unexpected machinery breakdowns remain a major concern, causing production downtime and significant financial losses. Therefore, there is a pressing need for an intelligent, integrated system that can address these challenges through automated monitoring, early risk detection, and predictive maintenance, ultimately enhancing both workplace safety and overall productivity.

1.2 Objectives of the Study

This study aims to:

1. Design an intelligent system based on Artificial Intelligence and IoT technologies to enhance workplace safety in factories.
2. Develop a mechanism to detect undesirable behaviors, such as workplace conflicts, and send instant alerts to security.
3. Ensure employee compliance with safety measures through automated detection of helmet usage.
4. Implement an attendance and departure system based on facial recognition to ensure accuracy and prevent manipulation.
5. Utilize sensor data to build a predictive model capable of forecasting potential machinery failures before they occur.
6. Integrate all these features into a unified platform accessible via both web and mobile applications for management and employees.

1.3 Graduation Project Organisation

The graduation project is organized into the following chapters:

- **Chapter One:** General introduction to the research topic and the significance of the project.
- **Chapter Two:** Literature review and related works on industrial safety systems and Artificial Intelligence.
- **Chapter Three:** Problem statement and formulation of the study objectives.
- **Chapter Four:** Methodology adopted in designing and developing the proposed system.
- **Chapter Five:** System description including architecture, technologies used, and component integration.
- **Chapter Six:** System implementation and testing with detailed results analysis.
- **Chapter Seven:** Conclusion, recommendations, and suggestions for future work.

References

Violence Detection

1. Research

- Violent Interaction Detection in Video Based on Deep Learning
<https://iopscience.iop.org/article/10.1088/1742-6596/844/1/012044>
- Advances in Information Communication Technology and Computing
Proceedings of AICTC 2021 (pp 165–173)
https://link.springer.com/book/10.1007/978-981-19-0619-0_2.

2. Dataset

- RLVS + Hockey Fight Dataset
<https://www.kaggle.com/datasets/inzilele/rlvs-hockey-fight-dataset-mp4-files>

3. Models

- CNN Feature Extractor → MobileNetV2 • Sequence Modeling → LSTM

Uniform/Non-Uniform Detection

1. Research

- Experiments in cross-domain few-shot learning for image classification <https://pmc.ncbi.nlm.nih.gov/articles/PMC11459747/>
- Automated uniform recognition to enhance video surveillance at correctional services in South Africa https://www.matecconferences.org/articles/mateconf/pdf/2023/15/mateconf_rapdasa2023_11001.pdf

2. Dataset

- People Clothing Segmentation (We will use a part of it) <https://www.kaggle.com/datasets/rajkumarl/people-clothing-segmentation>
- We will need to capture more photos of some working uniforms → about 100 photos

3. Models

- CNN Feature Extractor → MobileNetV2
- Cosine Similarity (Mathematical Metric) → Matching Logic.

Face Recognition

1. Research

- A Survey of Face Recognition

<https://arxiv.org/pdf/2212.13038>

2. Dataset

- LFW - People (For Benchmarking)

<https://www.kaggle.com/datasets/atulanandjha/lfwpeople>

- Data will be stored in DB for each employee in the working place

3. Models

- MTCNN (Multi-Task Cascaded CNN)
- ArcFace → Embedding = 512D • Cosine Similarity